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RESEARCH ARTICLE

Prioritizing Non-Functional Requirements in Agile Process Using Multi Criteria Decision Making Analysis

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ABSTRACT Agile software development does not include elicitation and management of non-functional requirements (NFRs). Several techniques have been proposed by researchers to elicit, manage and prioritize NFRs. The major issue with such techniques is that they involve customer or users to elicit and prioritize NFRs, which becomes problematic because they do not have exposure of NFRs. They may only be familiar with very few common NFRs such as performance, efficiency and security etc. There could be several other NFRs such as confidentiality, scalability, usability, maintainability, portability, reliability and many others which may not be known to the stakeholders. Consequently, some relevant NFRs may be missed and some wrong NFRs may be implemented in the final product which can degrade its overall quality. Secondly, it is very time consuming to elicit and prioritize NFRs from the stakeholders who have very low exposure. It is especially critical in the case of agile software development where product is developed and delivered at a very rapid pace. To overcome the challenge, this study proposed a model to prioritize NFRs for agile software development to produce quality software products. Thus, firstly, an extensive literature review was conducted to identify the NFRs which were further ranked through a survey carried out in agile software companies. Moreover, the NFRs were analyzed and prioritized using multi criteria decision making analysis technique called analytical hierarchical technique (AHP). The evaluation results showed that the NFRs prioritization model reduce the overhead of meetings and discussion required to gather and identify the NFRs common to agile products.

INDEX TERMS Agile software development, NFRs, requirement elicitation, requirements prioritization, software requirements.

I. INTRODUCTION

Traditional software development approaches have number of issues which are covered in the manifesto of agile methodologies [1]. Such approaches have some limitations to accommodate changes during software development whereas agile

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methodologies are more suitable and flexible to overcome the problem of frequently changing requirements [2]. A constant communication among stakeholders in agile improves customers' satisfaction and eventually product success rate [2]. It has been considered by several organizations that agile is suitable for handling large and complex problems [3], [4] and is best in producing software in highly volatile situations [5].

It has been reported by Standish group that agile projects are relatively more successful than projects executed through traditional approaches [6]. Agile methodology has become a dominant approach for software development as numerous organizations adopted it in past few years [1]. The requirement engineering process in software development relies on the heavy documentation for sharing domain knowledge among stakeholders but agile methods instead of documentation rely on face to face to communication between client and developers [7]. Agile is a people centric methodology rather than a process oriented approach, thus it is difficult to perform requirement work as performed in traditional software development through a proper requirement engineering processes [8]. In agile software development, requirement conflicts are mainly resolved through prioritization and negotiation with stakeholders [9].

Agile methodology does not rigorously and explicitly emphasize requirements; it concentrates on software development involving customer throughout the development life cycle. The focus during agile development process remains mainly on functional requirements and changes associated with such requirements. This scenario leads to the ignorance of NFRs. Both client and developer side stakeholders do not feel responsibility to define and implement NFRs. They think little about the beautification of interface and safety and neglect many important nonfunctional aspects of the system. Secondly as in agile there is no rigorous requirement engineering process to elicit, specify and prioritize requirement which further makes NFRs flawed. The flaws of NFRs may result into failure of software product because NFRs are the emergent properties of the software, represents the overall quality of the software product so these are more critical than functional requirements [10], [11]. In agile methodologies, there is a weaker treatment towards NFRs, they mainly focus on few most common NFRs such as usability, security and simplicity [12]. There are so many other NFRs including scalability, adaptability, stability, visibility, readability, volatility and accountability which need to be considered during agile software development.

One of the major reasons of neglecting NFRs in agile software development is the working philosophy of agile methods where product is implemented and handed over to a client through smaller iteration. During the process the developer and client concentrate on achieving the desired functionality and they ignore the NFRs. Another issue is the believe of agile practitioners regarding the importance of NFRs. For example, majority of the agile practitioners (i.e., 75%) don't consider that functional requirement and NFRs have equal importance whereas only 25% agile practitioners consider functional requirements and NFRs are of equal importance [13]. It is matter of paramount importance to contemplate about the NFRs at the initial stages of software development because overlooking them may lead to flaws, failure as well as increase in time, cost and maintenance [14].

There is a dearth of techniques in agile software development to prioritize both functional and non-functional requirements after elicitation. Later, the researchers have introduced various techniques including NORMAP, NERV, CEPP, CEP to elicit and prioritize NFRs in agile development. These techniques mainly based on user stories and training of client which take much time to process requirements.

This research study aimed to propose a ranking model to overcome the challenge of NFRs in agile development. The research process has started with the identification of significant NFRs from literature which have relevance to every agile software [15]. The identified NFRs were further filtered by a survey through companies practicing agile methodologies. The filtered NFRs have been ranked locally and globally through Analytical Hierarchy Process (AHP) [2]. The result is a prioritized list of the most significant NFRs that needs to be considered in every agile product.

The major purpose of this research is to prioritize NFRs from the perspective of agile development. The detailed objectives of this study are: (a) To identify common NFRs having impact on the quality of agile software product(s). (b) To classify NFRs according to their relative importance and impact. (c) To prioritize NFRs (locally and globally) according to their influence on quality of agile software product(s) as perceived by software organizations.

This study would be useful for agile software industry to overcome the challenges related to NFRs. It will reduce flaws in final product that occurs due to non-functional requirements. Furthermore, it will improve business and organizational quality at domestic and international scale.

The rest of this paper is organized as follows: Section II presents literature review. Section III describes research methodology. In section IV, we present and discuss the findings of this study and proposed a prioritization model along with its evaluation. Finally, in section V we give our conclusions and future work.

II. LITERATURE REVIEW

In recent years, agile process has turned into as an important methodology for software development mainly owing to its strength to accommodate changes across the development cycle. On the other hand, it has been observed that agile methods majorly focus on the functional aspects of the software and either neglect or poorly address the non-functional requirements of the system being under developed. However, for a good quality software both functional and non-functional aspects of the software should be taken into consideration. NFRs are the emerging properties of the software which represent the overall quality of the software. So the absence of adequate set of NFRs may lead to failure of a software product. Keeping in view, the importance of NFRs, researchers have greatly emphasized on their prioritization in agile software development. Suryawanshi & Rao proposed the design of a tool to model non-functional requirement for agile software development

process. The tool mainly elicits the requirements and link functional requirements with non-functional requirements [16]. Another research introduced a conceptual framework for non-functional requirements in agile software development process. The model linked the functional requirements with non-functional requirements and identify the riskiest requirements. The proposed framework is limited only to some 25 most critical non-functional requirements [17]. Domah & Mitropoulos proposed NERV methodology to effectively elicit the non-functional requirements in agile software development process along with the rationale of requirement selection and their verification [18]. Maiti & Mitropoulos introduced an approach to extract non-functional requirements from the metadata. The OCR technology has been used to translate metadata (i.e., images and pictures) into text document that is finally utilized to extract non-functional requirements [8]. The researchers further presented a mechanism to prioritize non-functional requirements using same approach used to prioritize functional requirements [19]. A research study explored that the user stories include information about quality attributes and identify such attributes using machine learning techniques. Few quality attributes or non-functional requirements were classified, however some of them found difficult to classify such as usability and reliability [20].

Another study proposed a methodology namely NFRElicit to gather non-functional requirements. The proposed methodology applied NLP based automatic non-functional extraction techniques to efficiently collect the requirements. It performs the role of experts and used the historical data of organization to elicit the non-functional requirements [21]. Jarzębowicz & Weichbroth investigated the issue of non-functional requirements in agile software development using qualitative research methods. They provide insight that there is no consensus of agile practitioners on the timing of eliciting NFRs. Furthermore, they have discovered that there is scarcity of techniques to elicit non-functional requirements for agile software projects [22]. A research study applied machine learning based automated technique to assess the security requirement of a software with regard to completeness, contradiction and inconsistency. Researchers used ISO standard to develop a model for the classification of security based requirements [23]. A study introduced an integrated approach to prioritize functional and non-functional stories in agile scrum software development method. They showed that there is a dire need to improve the prioritization process as it can help the product owner and other stakeholder in prioritizing user stories [24]. Another research introduced a fully integrated framework for agile processes to prioritize user stories at early stage of software development. The framework is particularly aimed at prioritization of functional requirements [25]. Veer & Singh used priority based algorithm to prioritize non-functional requirement in agile software development [26]. A web based decision making system based on ELECTRE as a multi criteria decision analysis

technique to prioritize functional requirements was proposed. The downside of this research is that the weightage of stakeholder was given by the product owner which may create biased ranking [27]. A study conducted to determine a most reliable method for security requirement engineering. This research used fuzzy TOPSIS multi-criteria decision making method to evaluate various security requirement engineering methods for the development of secure and trustworthy healthcare applications [28]. Another research conducted a longitudinal case study to provide insight regarding the priority assigned to security by security professionals in software development projects [29]. Dabbagh & Lee introduced an approach to prioritize both functional and non-functional requirement using decision matrix. The NFR which has high impact over functional requirement is selected as priority NFR. The proposed approach is only applicable to traditional software development [30]. For requirement prioritization a method called DRank has been proposed, it construct a tree of prioritization evaluation attributes for easier selection of ranking criteria. Furthermore, the method calculates the requirement prioritization considering stakeholder preferences and analyze the dependencies among requirement to modify the subjective preferences into objective requirement for rationale prioritization. The method is feasible to prioritize requirement in traditional software development [31]. Another study proposed a method to prioritize NFRs requirements initially independent from functional requirements and then their dependencies and usage counts with regard to functional requirement were identified using analytical hierarchical process [32]. Alawneh has introduced an identical approach for the prioritization of requirements by identifying relationships between functional and NFRs [33]. To deal with the issue of requirement prioritization specifically in the context component based software engineering a framework is proposed using text mining and case based reasoning [34]. Valsala proposed an algorithm to prioritize requirements for incremental development [35]. Another study proposed a hybrid model which consists of several techniques to prioritize requirements in scrum method of agile process. The authors expect that the proposed model will improve the overall quality of software system when it is included in scrum development process [36].

Previous research has introduced several methodologies, process and frameworks but still there is a need to establish a model to prioritize NFRs in agile projects. Determining and prioritizing NFRs especially in agile software development is a challenging process. Various techniques including binary search tree, spanning tree matrix, bubble sort, machine learning and visual tools have been utilized to prioritize requirements. However, these techniques did not take into consideration the perspective of domain experts while prioritizing NFRs. Requirement prioritization involves decision making that is an act of choosing the most significant from several requirements. Such decision making process

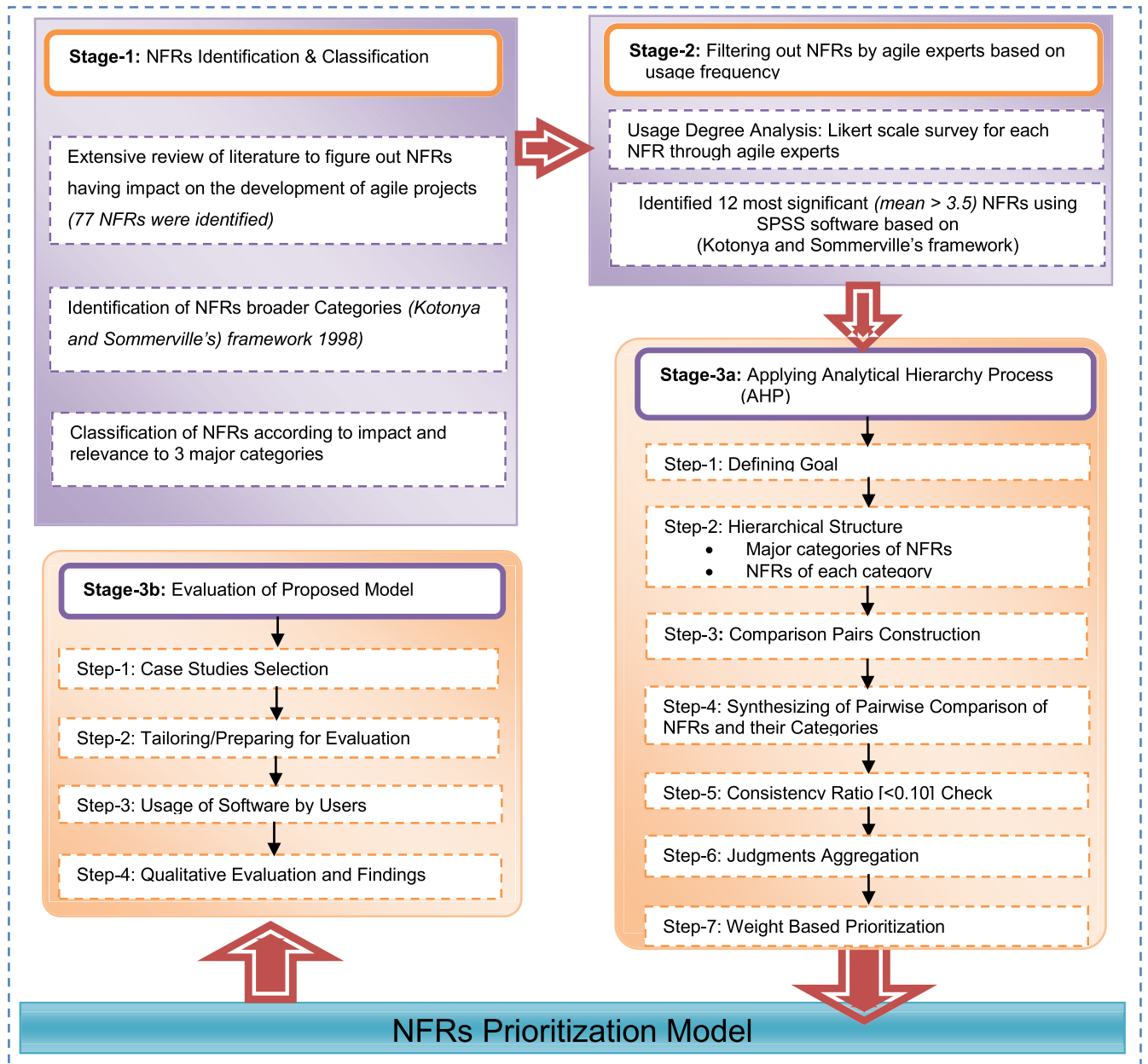


FIGURE 1. Graphical representation of research methodology.

requires the involvement of requirement engineering experts and appropriate technique(s).

Requirement prioritization is a multi-criteria decision making (MCDM) problem because diverse kind of NFRs are involved in the development of software systems so their significance can best be determined by MCDM. MCDM techniques have been widely applied in other areas, such as e-learning [37] usability [38], and e-commerce [39], [40] but, as per our knowledge such techniques are rarely explored for the prioritization of NFRs specifically in the context of agile software development. This research study is intended to present agile software developer's perspective on the relative

importance of the criteria for prioritizing NFRs model using multi criteria decision making methods (MCDM).

Various MCDM techniques have been discussed in the literature such as analytical hierarchy process (AHP), analytical network process (ANP), grey relation analysis (GRA), data envelopment analysis (DEA), preference ranking organization METHod for enrichment evaluation (PROMETHEE) and technique for order performance by similarity to ideal solution (TOPSIS) [38]. AHP is one of the best decision-making techniques of MCDM that is certainly good at making decisions through pairwise comparison of qualitative and quantitative factors. It offers an outstanding procedure to

calculate the weight of the criteria used in experts' reasoning processes [41], [42]. AHP is not only prioritize the requirements but also present the degree of importance of each requirement in relation to other requirements.

III. METHODOLOGY

The research methodology employed in this study comprises on three stages: in the first stage, an extensive literature review was conducted to explore and identify the non-functional aspects that impact the quality of agile software products. As a consequence of this in-depth literature review, the most appropriate NFRs were identified. In the next stage, a survey study was conducted to find out the usefulness of numerous identified non-functional aspects, and only those with a significant impact (key quality requirement) were further studies in this research. In final stage, the relative significance (weight) of each key non-functional requirement was calculated through the use of most widely applied pairwise comparison method—that is, the Analytical Hierarchical Process (AHP). The outcome of this stage is a ranked list of key quality requirements for agile software development systems, which is the underlying mechanism of our proposed quality requirements prioritization model. The graphical representation of the research methodology is shown in Figure 1.

The details of the three stages of research methodology are discussed below.

A. STAGE 1: IDENTIFY NON-FUNCTIONAL REQUIREMENTS

To identify the non-functional requirements for agile software products, an extensive literature review was conducted using a number of renowned sources including IEEE Explore (<https://ieeexplore.ieee.org>), Google Scholar (<https://scholar.google.com>), Science Direct (<https://www.sciencedirect.com>), Google Scholar (<https://scholar.google.com>), Springer (<https://www.springer.com>), ACM Digital Library (<https://www.acm.org>), ELSEVIER (<https://www.elsevier.com>), Research Gate (<https://researchgate.net>), Semantic Scholar (<https://www.semanticscholar.org>), Book Aid (<https://bookaid.org>). The result of literature showed the identification of a broad range of non-functional requirements, furthermore the identified non-functional requirements were classified into three dimensions in accordance to the framework introduced by Kotonya and Sommerville [43]. The three major dimensions include “Process”, “Product” and “External” requirements which are discussed in detail in section IV.

B. STAGE 2: DETERMINE THE IMPORTANCE OF NON-FUNCTIONAL REQUIREMENTS

As a result of stage 1, numerous non-functional requirements were identified. However, these non-functional requirements did not have similar importance with regard to agile software development. Therefore, the second stage aimed to conclude the significance of non-functional requirements in relation to agile software development, and only those

quality requirements which surpass the predetermined threshold were further taken into account. To define the key non-functional requirements, this research conducted a survey study as the survey method is an effective approach to collect data from a large number of participants. To test the reliability of self-designed questionnaire a pilot study was conducted using 15 participants. The coefficient value was 0.887 which indicate the reliability of questionnaire. To gather data, a Likert scale questionnaire was designed, it consists of a set of questions formulated to assess the significance of non-functional requirements identified in phase 1. The Likert scale questionnaire contains five options ranging from ‘very important’ to ‘not important’. The subjects of the survey were requested to state their opinion by selecting an appropriate option concerning each statement contributing to a specific dimension of quality requirements. To analyze the collected data a renowned statistical tool known as IBM SPSS (v.20) was used. The numeric values ranging from 1 to 5 were assigned to every option of the Likert scale, where 5 was allocated to very ‘important’ and 1 to ‘not important’. They survey targeted the agile software development practitioners who have at least 16 years of education in computer science, software engineering and information technology as well as at least two years of software development experience in agile environment. Total 392 participants were chosen mainly from the software industry of Pakistan. To choose an apposite sample, a non-probability sampling or snowball sampling technique was utilized. This technique commences with the identification of subjects possessed desired attributes. Later the social network of initially identified subjects is used to choose more participants. In this sampling technique, the size of a sample increases progressively similar to falling snow [38]. The selected sample comprised on 392 subjects. The demographic profile of respondents was as follows. The sample contains both male and female subjects. The male subjects were 243 (62%) and females were 149 (38%). In terms of qualification 314 (80%) were with 16 years of education and rest of the 78 (20%) subjects were with 18 years of education in computing disciplines (i.e., computer science, software engineering and information technology).

C. STAGE 3: PRIORTIZING KEY NON-FUNCTIONAL REQUIREMENTS USING AHP

The major objective of this stage was to determine the relative weights and ranking of the key quality requirements identified during in previous stage 2. The pairwise comparison method was used to assess the weights of the dimensions of non-functional requirements. A pairwise comparison is a process to equate objects in the form of pairs to judge which objects is being preferred by experts or showing a greater amount of quantitative property or whether or not two objects have similarity. It originated by analytical hierarchical process (AHP) [44] a widely known multi criteria decision making method which is applied in scientific study of preferences, ecommerce and eLearning. This study employed the

pairwise comparison approach [45] to assign weights to the key quality requirements as discussed below.

Step 1: The pairwise comparison matrix was accomplished by evaluation two key requirements at a time with respect to their relative importance through index values from 1 to 9. If requirement R_i has the similar importance as requirement R_j , this pair obtained an index of 1. On the other hand, if R_i was exceptionally important than R_j , the index was 9. The range of possible gradation is shown in Table 1. For an 'insignificant' relationship, the fractions 1/1 to 1/9 were available: if R_i was extraordinarily insignificant than R_j , the grading was 1/9.

The values of index were evaluated based on consensus by domain experts, for this purpose, a questionnaire was designed and spread among the participants who had at least two years of experience in agile software development environment. The participants were constrained with the amount of experience because a certain years of experience in agile development environment could enable them to take right decision regarding the significance of NFRs. The experts were asked to assess the significance of each key quality requirement in relation to other key quality requirements and record their evaluations, as shown in Table 1. If the experts varied in terms of their evaluations, consensus approach was used to reduce discrepancy. The ultimate accepted evaluations were recorded row wise into a cross-matrix C ($n \times n$). At the outset, the diagonal of C was recorded by values of 1 in accordance to (1). Afterward, the right upper half of C was recorded until each key quality requirement had been compared to every other one. If R_i to R_j were rated with the relative significance of m (i.e., $C_{ij} = m$), R_j to R_i have to be rated with $1/m$ (i.e., $C_{ji} = 1/m$). Finally, the lower left half of C was filled with the corresponding fractions in accordance to (2). (Note that i and j are positive integers $\leq n$, C_{ij} is the part of C placed in row i and column j).

$$C_{ij} = 1, \quad i = j \quad (1)$$

$$C_{ij} = \frac{1}{C_{ji}}, \quad i \neq j \quad (2)$$

Step 2: the normalized comparison matrix was calculated. A normalized comparison matrix C' was formed by dividing each component in matrix C by the summation of the components in its column. This is shown using (3).

$$C'_{ij} = \frac{C_{ij}}{\sum_{i=1}^n C_{ij}} \quad (3)$$

Step 3: the relative weights of the key quality requirements were calculated. To obtain the weight w_i of each individual quality requirement R_i , the mean of every row in C' was calculated, as shown in (4).

$$w_i = \frac{1}{n} \sum_{j=1}^n C'_{ij} \quad (4)$$

These weights were already normalized; their sum was 1, as shown in (5) and (6).

$$0 \leq w_i \leq 1 \quad (5)$$

$$\sum_{i=1}^n w_i = 1 \quad (6)$$

Step 4: at this step, the consistency of the results of the pairwise comparisons were verified. As per Saaty, if the consistency ratio was below than 10%, then it was satisfactory; otherwise, the pairwise comparisons need to be revised [46]. The ratio of consistency (CR) is illustrated in (7).

$$CR = \frac{CI}{RI} \quad (7)$$

where CI is consistency index and is illustrated by Equation 8.

$$CI = \frac{\lambda_{max} - n}{n - 1} \quad (8)$$

λ_{max} is the extreme eigenvalue and n is the rank of the pairwise comparison matrix [37].

TABLE 1. Index values in pairwise comparison matrix AHP scale.

Definition	Index	Definition	Index
Equally important	1	Equally important	1/1
Moderately more important	3	Moderately less important	1/3
Much more important	5	Much less important	1/5
Far more important	7	Far less important	1/7
Almost always/Extremely more important	9	Extremely less important	1/9

RI specifies the random index of consistency, which had a different value based on the number of criteria, as shown in Table 2.

TABLE 2. Relationship between RI and N.

N	1	2	3	4	5	6	7	8	9	10
RI	0	0	0.58	0.9	1.12	1.24	1.32	1.41	1.45	1.49

Various software applications such as Expert Choice, MS Excel, XLSTAT, AUTOMAN, HIPRE3 and Criterium etc., are being used by researchers to implement AHP method. However, the Expert Choice is known as the standard AHP software [47], that is why it was used in this study along with the support of MS Excel. In this research, AHP was implemented using following steps. (a) The objective of the research was defined, which was to estimate the importance of NFRs in the context of agile software development.

(b) The research objective and evaluation criteria were organized in a hierarchical structure where first level of hierarchy define the objective, second level define the main quality requirements criteria, and the third level define the sub-criteria related to each main criterion. (c) The relative weights for criteria and sub-criteria were computed through

steps from 1 to 4, as discussed above. A pairwise comparison requires $n(n-1)/2$ comparison [46], where 'n' represents the number of criteria or sub-criteria.

IV. RESULT AND DISCUSSION

The findings of three stages explained in previous section are discussed below.

A. FINDINGS OF STAGE-1: LITERATURE REVIEW

The literature review stage 1 resulted in the identification of a large number of useful NFRs in the context of agile software development. These NFRs were deeply studied and classified into three major dimensions including external, process and product requirements. The major dimensions are also known as global non-functional requirements. External requirements majorly contain all those requirements which influence the final software product or product development procedure externally such as interoperability, ethical and Legislative requirements etc. Process or organizational requirements are those requirements that are inherited or resulting from the rules, strategies and actions of an organization. Product requirements are those requirements which specify that how software product is being expected to perform in terms of operations, efficiency and reliability. Table 3 presents the NFRs along with sources of extraction.

Numerous NFRs as shown in above Table 3 have been found as a result of extensive review of literature pertinent to agile software development. However, some NFRs were discarded on the basis of frequency of citation in agile software development literature. The NFRs which appeared in agile literature only once or twice were discarded considering them unimportant. So only those NFRs were considered for stage 2 which frequently cited in the agile literature.

B. FINDINGS OF STAGE 2

In second stage, a survey was conducted to judge the importance of each non-functional requirement on a 5 point Likert scale. The survey was conducted using Google form which provides the clear and concise definition of each NFR so that the subjects of the survey could understand it well before answering. The objective the study and importance of their responses with respect to quality of agile projects was described in clear terms. The questionnaire was distributed among a sample of 392 subjects who were brought into reach through telephonic conversation and face to face meetings upon their convenient availability. However, out of 392 subjects only 186 (47%) has responded. The details of the respondents are depicted in below Table 4.

All of the participants of this survey were agile experts with minimum 2 years' experience in agile software development and having maximum experience more than 10 years. Majority of the participants (88%) have 2-5 years' experience in agile software development whereas (12%) participants have around 10 years or more than 10 years' experience in agile development. The responses of agile experts obtained through questionnaire and analysed using IBM statistical

TABLE 3. Extracted non-functional requirements.

Major Dimensions of NFRs	NFRs	References
External	Accessibility	[8] [48] [49] [51]
	Anonymity	[52]
	Authentication	[49] [52]
	Automaticity	[53]
	Branding	[54]
	Changeability	[48]
	Compatibility	[51] [52] [54]
	Confidentiality	[5] [8] [51] [52]
	Integrity	[52]
	Internationalization	[66]
	Interoperability	[8] [51] [52]
	Legal	[8] [51] [52]
	Multilingual	[8] [51] [52]
	Pseudonymity	[52]
	Privacy	[49] [52]
	Quality	[3] [5] [48] [54] [63] [65] [66]
	Safety	[16] [52] [54] [57] [60]
	Security	[3] [8] [14] [48] [51] [52] [54] [55] [56] [57] [58] [59] [60] [61]
	Significantly	[62]
	Suitability	[5] [48] [54] [63]
Process/Organizational	Tailorability	[52]
	Value	[54]
	Volatility	[60] [63] [64]
	Acceptance	[48] [65] [66]
	Accountability	[64]
	Awareness	[52]
	Commonality	[63]
	Communication	[56] [67]
	Constraints	[48] [56]

software package SPSS software (v.20). the results are shown in Table 5.

TABLE 3. (Continued.) Extracted non-functional requirements.

Product	Cost-affectivity	[3]
	Documentation	[8] [51]
	Feasibility	[2] [3] [48]
	Flexibility	[2] [48] [54] [57] [59]
	Health	[54]
	Industry Standards	[48]
	Inspect-ability	[2]
	Maintenance	[14]
	Modifiability	[2]
	Predictability	[2]
	Scalability	[8] [48] [51] [55] [60] [61] [64] [66]
	Supportability	[49]
	Testability	[2] [14] [53]
	Traceability	[59] [65]
	Transparency	[2] [53] [64] [66]
	Accuracy	[8] [51] [54]
	Adaptability	[2] [52] [54] [64] [65]
	Affectivity	[2] [54] [65]
	Availability	[8] [48] [51] [52] [55]
	Capacity/capability	[52]
	Compliance	[8] [51]
	Complexity	[52]
	Configuration	[8] [51]
	Content	[54]
	Consistency	[5] [55]
	Dependability	[56]
	Design (UCD)	[12] [48] [54]
	Desirability	[54]
	Efficiency	[2] [5] [8] [48] [51] [54] [65]
	Functionality	[48] [54]

The below Table 5 shows the analysis of data related to NFRs of external category collected by agile experts through

TABLE 3. (Continued.) Extracted non-functional requirements.

Interface/Content Structure	[1] [8] [51] [54] [56]
Learnability	[2] [54]
Maintainability	[1] [5] [60] [66]
Operability	[66]
Performance	[1] [5] [8] [14] [48] [49] [50] [51] [52] [54] [56] [57] [58] [59] [60] [61] [62] [66]
Portability	[1] [2] [48] [60]
Productivity	[2]
Readability	[64]
Reliability	[2] [5] [8] [48] [49] [51] [54] [55] [57] [66]
Responsibility	[53]
Satisfaction	[54]
Simplicity	[2] [48] [52]
Stability	[2] [61] [64]
Structure	[48]
Trust-ability	[52]
Understandability	[2] [48] [49]
Usability	[5] [8] [12] [48] [49] [51] [54] [55] [57] [59] [60] [62] [66]
Variability	[63] [66]

TABLE 4. Details of participants.

Qualificati on	Frequen cy	%age	Experien ce	Frequen cy	%age
BS in computing	155	83.33 %	2-5 yrs.	164	88.17 %
MS in computing	25	13.44 %	6-10 yrs.	20	10.75 %
PhD in computing	6	3.23%	>10 yrs.	02	1.08%

a survey. The graphical representation of these NFRs is shown in Figure 2 (Appendix A).

The finding of stage 2 about “external” NFRs shows that accessibility, compatibility, quality and security are the 4 out of 11 external requirements having mean value greater than

TABLE 5. Analysis of data about external NFRs.

NFRs	Scale	Degree of Usage					Mean	Std. Deviation
		Never	Seldom	Occasionally	Often	Almost Always		
Accessibility	186	11	20	26	60	69	3.84	1.20
	%	05.91	10.75	13.97	32.25	37.09		
Compatibility	186	12	31	39	47	57	3.57	1.26
	%	06.45	16.66	20.96	25.26	30.64		
Confidentiality	186	28	43	31	33	51	3.19	1.44
	%	15.05	23.11	16.66	17.74	27.41		
Interoperability	186	31	51	25	46	33	2.99	1.38
	%	16.66	27.41	13.44	24.73	17.74		
Legal	186	35	37	36	28	50	3.11	1.47
	%	18.81	19.89	19.35	15.05	26.88		
Multilingual	186	53	33	30	36	34	2.81	1.49
	%	28.49	17.74	16.12	19.35	18.27		
Quality	186	10	21	25	53	77	3.89	1.21
	%	05.37	11.29	13.44	28.49	41.39		
Safety	186	24	32	26	37	67	3.49	1.45
	%	12.90	17.20	13.97	19.89	36.02		
Security	186	12	12	31	62	69	3.88	1.17
	%	06.45	06.45	16.66	33.33	37.09		
Suitability	186	29	36	30	29	62	3.32	1.49
	%	15.59	19.35	16.12	15.59	33.33		
Volatility	186	43	32	30	34	47	3.05	1.51
	%	23.11	17.20	16.12	18.27	25.26		

3.5, as shown in Table 5. The ‘quality’ is found as the most important NFR in agile software projects having relatively greater mean value (i.e., 3.89) whereas ‘multilingual’ is with least mean value (i.e., 2.81).

The below Table 6 shows the analysis of data related to NFRs of “organizational/process” category collected by agile experts through a survey study. The graphical representation of these NFRs is shown in Figure 3 (Appendix A).

The finding of stage 2 about “organizational/process” NFRs shows acceptance, feasibility and scalability are the 3 out of 8 organizational requirements having mean value greater than 3.5, as shown in Table 6. The ‘acceptance’ is found as the most important NFR in agile software projects having relatively greater mean value (i.e., 3.88) whereas ‘constraints’ is found an organizational requirement having least mean value (i.e., 2.97).

TABLE 6. Analysis of organizational/process NFRs.

NFRs	Scale	Degree of Usage					Mean	Std. Deviation
		Never	Seldom	Occasionally	Often	Almost Always		
Acceptance	N	10	13	33	64	66	3.88	1.13
	%	05.37	06.98	17.74	34.40	35.48		
Communication	N	34	33	33	48	38	3.12	1.40
	%	18.27	17.74	17.74	25.80	20.43		
Constraints	N	36	53	21	33	43	2.97	1.47
	%	19.35	28.49	11.29	17.74	23.11		
Feasibility	N	17	17	29	67	56	3.69	1.24
	%	09.13	09.13	15.59	36.02	30.10		
Flexibility	N	33	44	21	42	46	3.13	1.46
	%	17.74	23.65	11.29	22.58	24.73		
Scalability	N	18	22	29	61	56	3.62	1.29
	%	09.67	11.82	15.59	32.79	30.10		
Testability	N	38	42	17	40	49	3.11	1.52
	%	20.43	22.58	09.13	21.50	26.34		
Transparency	N	41	28	27	44	46	3.14	1.50
	%	22.04	15.05	14.51	23.65	24.73		

The below Table 7 shows the analysis of data related to NFRs of “product” category collected by agile experts through a survey study. The graphical representation of these NFRs is shown in Figure 4 (Appendix A).

The finding of stage 2 about “product” NFRs shows accuracy, efficiency, interface, performance and stability are the 5 out of 15 product requirements having mean value greater than 3.5, as shown in Table 7. The survey results

TABLE 7. Analysis of product NFRs.

NFRs	Scale	Degree of Usage					Mean	Standard Deviation
		Never	Seldom	Occasionally	Often	Almost Always		
Accuracy	N	12	07	18	64	85	4.09	1.13
	%	06.45	03.76	09.67	34.40	45.69		
Adaptability	N	22	50	35	39	40	3.13	1.34
	%	11.82	26.88	18.81	20.96	21.50		
Affectivity	N	30	40	33	38	45	3.15	1.42
	%	16.12	21.50	17.74	20.43	24.19		
Availability	N	27	31	37	43	48	3.29	1.39
	%	14.51	16.66	19.89	23.11	25.80		
Design (UCD)	N	22	44	24	44	52	3.32	1.40
	%	11.82	23.65	12.90	23.65	27.95		
Efficiency	N	08	07	31	65	75	4.03	1.05
	%	04.30	03.76	16.66	34.94	40.32		
Interface/Content Structure	N	08	19	21	69	69	3.92	1.13
	%	04.30	10.21	11.29	37.09	37.09		
Maintainability	N	20	52	16	48	50	3.30	1.40
	%	10.75	27.95	08.60	25.80	26.88		
Performance	N	09	14	25	60	78	3.99	1.14
	%	04.83	07.52	13.44	32.25	41.93		
Portability	N	39	41	20	33	33	3.11	1.54
	%	20.96	22.04	10.75	17.74	28.49		
	N	29	34	21	55	47		

revealed that ‘accuracy’ is relatively the most important NFR in agile software projects having relatively greater mean value

TABLE 7. (Continued.) Analysis of product NFRs.

Reliability	%	15.59	18.27	11.29	29.56	25.26	3.31	1.42
	N	21	30	30	45	60	3.50	1.38
Simplicity	%	11.29	16.12	16.12	24.19	32.25		
	N	10	17	35	61	63	3.81	1.16
Stability	%	05.37	09.13	18.81	32.79	33.87		
	N	29	44	23	38	52	3.22	1.46
Usability	%	15.59	23.65	12.36	20.43	27.95		
	N	38	29	17	38	64	3.33	1.56
Understandability	%	20.43	15.59	09.13	20.43	34.40		

(i.e., 4.09) whereas portability is having least mean value (i.e., 2.11).

The below Table 8 shows the analysis of three major categories of NFRs including “process”, “organization/process” and “product” based on the data collected by agile experts through a survey study. The graphical representation of the global categories of NFRs is shown in Figure 5 (Appendix A).

The findings show that all three global categories of NFRs have mean value greater than 3.5, as shown in Table 8. It has been shown that “product” category of NFRs is relatively more important with 4.5 mean value whereas “External” category of NFRs has least mean value (i.e., 4.2.).

Table 9 presents an integrated analysis of all NFRs of three major categories including “External”, “Process” and “Product” in terms of high to low mean value of each NFRs. The graphical representation of all NFRs of three major categories is shown in Figure 6 (Appendix A).

As the goal of MCDM methods is to compare limited factors with each other to reach a decision point. However extensive review of literature produces a large number of elements which need to be reduced so that the comparison using MCDM method (i.e., AHP) may performed through domain experts. It is therefore there is a need to specify cutoff point to discard some less important or infrequently used factors (NFRs in context of this study). The requirements having mean value below 3.5 has been discarded and not made part of further analysis. The less important requirements were discarded using threshold value/cutoff that was chosen on

the basis of experts and researchers choice. After reviewing all of the responses and consulting with experts in the area, mean value 3.5 was set as most apposite threshold value, as if we set less than 3.5 vary few requirements were likely to be discarded and if set it greater than 3.5 then most significant requirements were expected to be discarded.

The above Table 9 shows that 12 out of 34 NFRs of three major categories have mean value greater than 3.5. Such requirements include accessibility, compatibility, quality, security, acceptance, feasibility, scalability, accuracy, efficiency, interface, performance and stability.

The results in Table 9 demonstrates that the ‘accuracy’ has been recognized by most of the respondents (80%) as a non-functional requirement that is considered most frequently in agile software projects. The mean value of ‘accuracy’ is 4.09 that is highest among all non-functional requirements of three categories. The ‘efficiency’ is a second non-functional requirement which has been identified by most of the respondents (75%), the mean value of ‘efficiency’ is 4.03. Other NFRs such as ‘performance’ and ‘interface’ are perceived by most of the respondents (74%) with mean value 3.99 and 3.92 correspondingly. On the other hand, interoperability, constraints (e.g. weight, cultural) and multilingual have been considered by more than 60% respondents as the NFRs which used rarely in agile projects. Such requirements have lowest mean value that is (2.99), (2.97), (2.81) respectively.

TABLE 8. Analysis of global categories of NFRs.

Major Categories of NFRs	Scale	Degree of Usage					Mean	Std. Deviation
		Never	Seldom	Occasionally	Often	Almost Always		
External NFRs	N	03	06	23	72	82	4.20	0.89
	%	01.61	03.22	12.36	38.70	44.08		
Process NFRs	N	03	03	14	57	109	4.43	0.83
	%	01.61	01.61	07.52	30.64	58.60		
Product NFRs	N	03	02	12	51	118	4.5	0.80
	%	01.61	01.07	06.45	27.41	63.44		

C. FINDINGS OF STAGE-3: AHP IMPLEMENTATION

In this stage AHP method was applied to estimate the relative weights and to rank the NFRs and their major categories. A questionnaire was designed in the form of a pairwise comparison based on the hierarchical structure.

TABLE 9. Analysis of NFRs of three major categories.

NFRs	Scale	Degree of Usage					Mean	Std. Deviation
		Never	Seldom	Occasionally	Often	Almost Always		
Accessibility	N	11	20	26	60	69	3.84	1.20
	%	05.91	10.75	13.97	32.25	37.09		
Compatibility	N	12	31	39	47	57	3.57	1.26
	%	06.45	16.66	20.96	25.26	30.64		
Confidentiality	N	28	43	31	33	51	3.19	1.44
	%	15.05	23.11	16.66	17.74	27.41		
Interoperability	N	31	51	25	46	33	2.99	1.38
	%	16.66	27.41	13.44	24.73	17.74		
Legal	N	35	37	36	28	50	3.11	1.47
	%	18.81	19.89	19.35	15.05	26.88		
Multilingual	N	53	33	30	36	34	2.81	1.49
	%	28.49	17.74	16.12	19.35	18.27		
Quality	N	10	21	25	53	77	3.89	1.21
	%	05.37	11.29	13.44	28.49	41.39		
Safety	N	24	32	26	37	67	3.49	1.45
	%	12.90	17.20	13.97	19.89	36.02		
Security	N	12	12	31	62	69	3.88	1.17
	%	06.45	06.45	16.66	33.33	37.09		

As discussed earlier after cutoff the 22 NFRs were discarded and only 12 were considered for further analysis.

TABLE 9. (Continued.) Analysis of NFRs of three major categories.

Suitability	N	29	36	30	29	62	3.32	1.49
	%	15.59	19.35	16.12	15.59	33.33		
Volatility	N	43	32	30	34	47	3.05	1.51
	%	23.11	17.20	16.12	18.27	25.26		
Acceptance	N	10	13	33	64	66	3.88	1.13
	%	05.37	06.98	17.74	34.40	35.48		
Communication	N	34	33	33	48	38	3.12	1.40
	%	18.27	17.74	17.74	25.80	20.43		
Constraints	N	36	53	21	33	43	2.97	1.47
	%	19.35	28.49	11.29	17.74	23.11		
Feasibility	N	17	17	29	67	56	3.69	1.24
	%	09.13	09.13	15.59	36.02	30.10		
Flexibility	N	33	44	21	42	46	3.13	1.46
	%	17.74	23.65	11.29	22.58	24.73		
Scalability	N	18	22	29	61	56	3.62	1.29
	%	09.67	11.82	15.59	32.79	30.10		
Testability	N	38	42	17	40	49	3.11	1.52
	%	20.43	22.58	09.13	21.50	26.34		
Transparency	N	41	28	27	44	46	3.14	1.50
	%	22.04	15.05	14.51	23.65	24.73		

So the AHP questionnaire was designed in accordance to the qualifying 12 NFRs. The questionnaire was distributed among 28 agile experts of four different software houses who have 18 years’ qualification in computing programs

TABLE 9. (Continued.) Analysis of NFRs of three major categories.

Accuracy	N	12	07	18	64	85	4.09	1.13
	%	06.45	03.76	09.67	34.40	45.69		
Adaptability	N	22	50	35	39	40	3.13	1.34
	%	11.82	26.88	18.81	20.96	21.50		
Affectivity	N	30	40	33	38	45	3.15	1.42
	%	16.12	21.50	17.74	20.43	24.19		
Availability	N	27	31	37	43	48	3.29	1.39
	%	14.51	16.66	19.89	23.11	25.80		
Design (UCD)	N	22	44	24	44	52	3.32	1.40
	%	11.82	23.65	12.90	23.65	27.95		
Efficiency	N	08	07	31	65	75	4.03	1.05
	%	04.30	03.76	16.66	34.94	40.32		
Interface /Content Structure	N	08	19	21	69	69	3.92	1.13
	%	04.30	10.21	11.29	37.09	37.09		
Maintainability	N	20	52	16	48	50	3.30	1.40
	%	10.75	27.95	08.60	25.80	26.88		
Performance	N	09	14	25	60	78	3.99	1.14
	%	04.83	07.52	13.44	32.25	41.93		
Portability	N	39	41	20	33	53	3.11	1.54
	%	20.96	22.04	10.75	17.74	28.49		

(e.g. computer science, software engineering, information technology and software project management) and at least six years’ experience in agile software development. The agile

TABLE 9. (Continued.) Analysis of NFRs of three major categories.

Reliability	N	29	34	21	55	47	3.31	1.42
	%	15.59	18.27	11.29	29.56	25.26		
Simplicity	N	21	30	30	45	60	3.50	1.38
	%	11.29	16.12	16.12	24.19	32.25		
Stability	N	10	17	35	61	63	3.81	1.16
	%	05.37	09.13	18.81	32.79	33.87		
Usability	N	29	44	23	38	52	3.22	1.46
	%	15.59	23.65	12.36	20.43	27.95		
Unders tandability	N	38	29	17	38	64	3.33	1.56
	%	20.43	15.59	09.13	20.43	34.40		

experts were approached personally through visiting their workplaces. Among 28 subjects only 12 participants returned the questionnaires.

AHP method has several steps which are implemented by three different tools including Expert Choice software, Microsoft Excel and AHP online software. The steps of implementation along with corresponding findings are discussed below.

1) DEFINE GOAL, CATEGORIES AND NFRs TO ESTABLISH HIERARCHAL MODEL

The hierarchical model was created by defining the goal (*i.e., prioritization of non-functional requirements in agile software development*), global categories of NFRs and requirements related to these categories chosen through previously discussed process. The hierarchical model is shown in Figure 7 which demonstrates the main goal of this research, three major categories of NFRs namely “external”, “process” and “product”. The selected 12 requirements are defined beneath each category of NFRs.

2) CONSTRUCTING PAIRWISE COMPARISON MATRICES/JUDGMENT MATRICES BASED ON THE DATA COLLECTED BY DOMAIN EXPERTS

After constructing hierarchical structure, the contribution of each NFR and global categories were calculated. In this step, pairwise comparison have been made as discussed in

step 1 to 4 of sub section III-C. The primary concept of AHP is pairwise comparison. The most important step here is step 1 where experts develop consensus based on the index values entered to pairwise comparison matrices. According to AHP procedure, a pairwise judgement matrix for each decision maker is constructed.

In AHP questionnaire, 16 out of 28 survey responses were discarded as their consistency ratio was greater than 10%. So only remaining 12 responses were used for detailed analysis, these responses present relative score for each NFR and are placed into judgment matrix. Table 10 presents the judgment matrix containing relative score from the all 12 participants for three global categories of NFRs including “External”, “Process” and “Product”. Table 11 presents the judgment matrix containing relative score from the all 12 participants for NFRs of “external” category. Similarly, Table 12 and 13 present judgment matrix for “process” and “product” category of NFRs.

3) PERFORMING CONSISTENCY RATIO ON EXPERTS’ INPUT AND CALCULATING PRIORITY WEIGHTS OF EACH MAJOR CATEGORY AND CORRESPONDING NFRs

After founding the judgment matrices, the importance or priority weights of major categories of NFRs (aka global categories of NFRs) were calculated. This process is technically known as eigenvector which refers to the relative contribution of one element over other. Finalizing this process involves consistency testing of each participant judgments. Consistency ratio (CR) testing is an important process that heavily affects AHP that is illustrated in step 4 of sub-section C of methodology. Table 14 (Appendix A) shows the consistency index (CI), $\lambda_{\max}$ (maximum Eigen value), random index (RI) and CR computed based on the responses given by agile experts for major categories of NFRs including “External”, “Process” and “Product” requirements. The elicited responses of experts were fall within the acceptable range of CR.

The same process has been applied to calculate the weights of NFRs of each major category of requirements based on the responses of agile experts. Table 15 (Appendix A) shows the consistency index (CI), $\lambda_{\max}$ (maximum Eigen value), random index (RI) and CR for ‘External’ NFRs computed based on the responses provided by agile experts with acceptable CR. Table 16 (Appendix A) shows the consistency index (CI), $\lambda_{\max}$ (maximum Eigen value), random index (RI) and CR for ‘Process’ NFRs computed based on the responses given by agile experts with acceptable CR. Table 17 (Appendix A) shows the consistency index (CI), $\lambda_{\max}$ (maximum Eigen value), random index (RI) and CR for ‘Process’ NFRs computed based on the responses specified by agile experts fall within the range of acceptable CR.

4) AGGREGATIONS OF JUDGEMENT

Judgments aggregation process has been used to aggregate the values provided by the experts in the form of responses

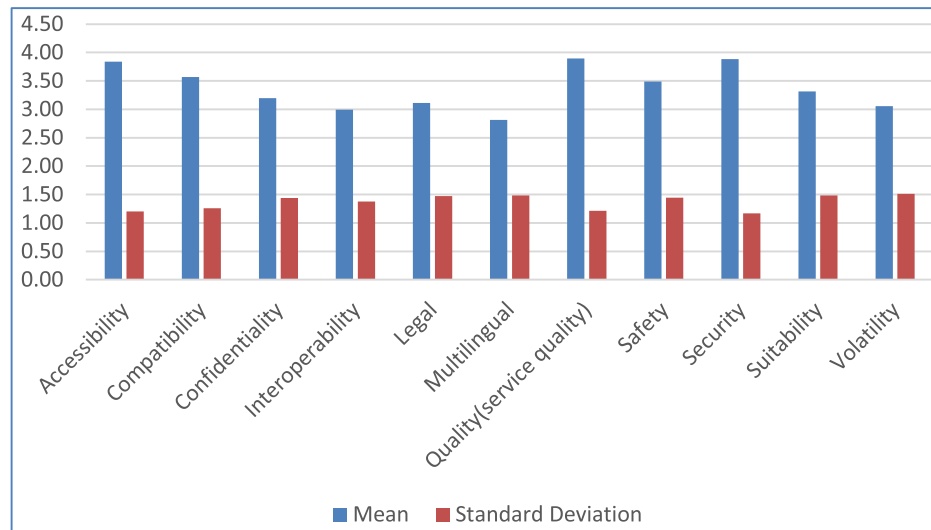


FIGURE 2. Graphical representation of External category NFRs.

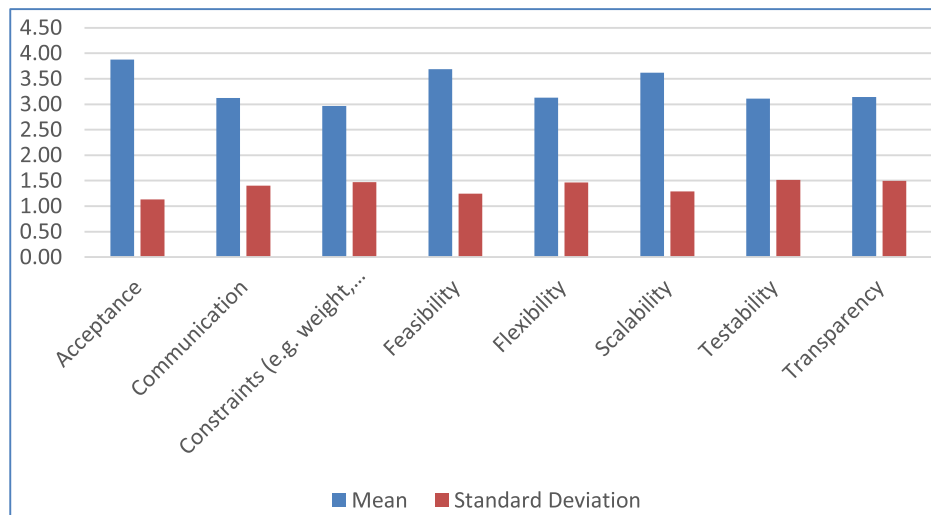


FIGURE 3. Graphical representation of Process category NFRs.

with respect to NFRs in agile software development. The aggregation process is applied to combine the several values against each major category and corresponding NFRs to form a single value matrix. The aggregation process has been completed through Expert Choice software by combining the responses from all experts. Table 18 shows the aggregated priority weights of major categories of NFRs, the “Product” category of NFRs has been found as the most important category with highest aggregated priority weight i.e., 0.394. The “External” category of NFRs has been found discovered as the second most important category of NFRs with aggregated priority weight i.e., 0.393. However, there is slightest difference between the priority weights of “External” and “Product” categories of NFRs which indicate that both categories of NFRs are equally used in the context of agile software development. The priority weight of “Process” category of

NFRs 0.213 that is low as compare to other two categories of NFRs.

The same process has been applied to calculate the weights of NFRs of all categories including “External”, “Process” and “Product” based on the responses of each expert. The below Table 19 shows the aggregated priority weights of NFRs of each category. In “External” category, ‘accessibility’ was relatively found high priority weighted NFR. Similarly, ‘acceptance’ found high priority weighted NFR in “Process” and ‘accuracy’ in “Product” category.

5) RANKING THE REQUIREMENTS BASED ON THEIR PRIORITIES

Requirements ranked based on their priority weights which are further divided into local and global weights. Local

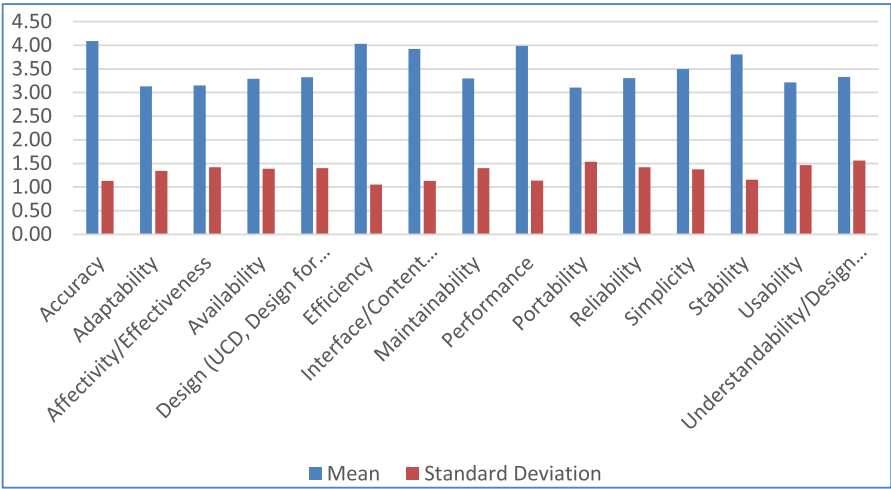


FIGURE 4. Graphical representation of Process category NFRs.

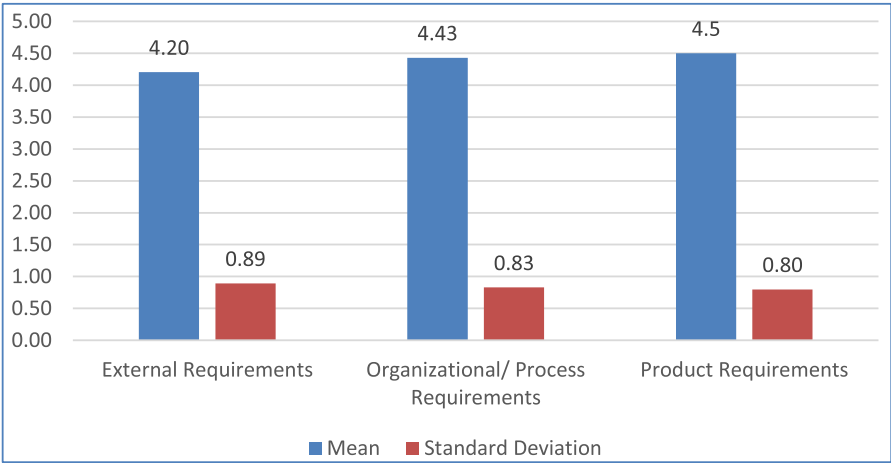


FIGURE 5. Graphical representation of major categories of requirements.

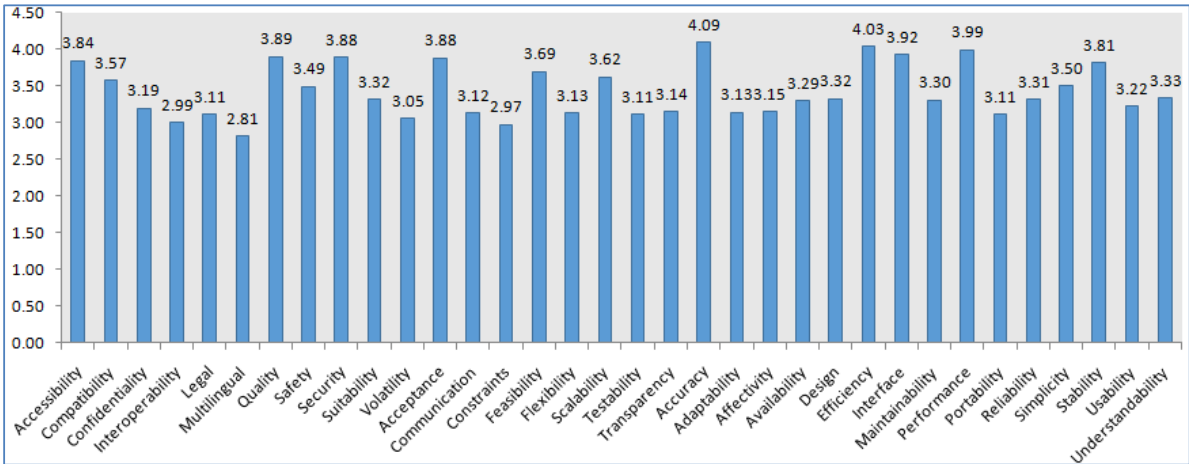


FIGURE 6. Graphical representation of all NFRs of three major categories.

refers to the priority weights for preceding hierarchical level whereas global refers to priority weights for highest hierarchical level. AHP combines the priority weights of categories with the comparison ratings requirements for this process of local and global ranking. The below Table 20 shows the overall weights and rankings (locally as well as

TABLE 10. Judgement matrix of global categories of NFRs based on each expert response.

	External NFRs	Process NFRs	Product NFRs
1 st Expert	External NFRs	1.00	4.00
	Process NFRs	0.25	1.00
	Product NFRs	0.50	3.00
2 nd Expert	External NFRs	1.00	4.00
	Process NFRs	0.25	1.00
	Product NFRs	0.20	0.50
3 rd Expert	External NFRs	1.00	2.00
	Process NFRs	0.50	1.00
	Product NFRs	3.00	5.00
4 th Expert	External NFRs	1.00	7.00
	Process NFRs	0.14	1.00
	Product NFRs	0.20	3.00
5 th Expert	External NFRs	1.00	5.00
	Process NFRs	0.20	1.00
	Product NFRs	3.00	9.00
6 th Expert	External NFRs	1.00	7.00
	Process NFRs	0.14	1.00
	Product NFRs	0.33	5.00
7 th Expert	External NFRs	1.00	3.00
	Process NFRs	0.33	1.00
	Product NFRs	0.25	1.00
8 th Expert	External NFRs	1.00	3.00
	Process NFRs	0.33	1.00
	Product NFRs	0.25	0.33
9 th Expert	External NFRs	1.00	0.20
	Process NFRs	5.00	1.00
	Product NFRs	2.00	0.20
10 th Expert	External NFRs	1.00	5.00
	Process NFRs	0.20	1.00
	Product NFRs	3.00	7.00
11 th Expert	External NFRs	1.00	4.00
	Process NFRs	0.25	1.00
	Product NFRs	0.50	1.00
12 th Expert	External NFRs	1.00	0.33
	Process NFRs	3.00	1.00
	Product NFRs	4.00	2.00

globally) of requirement categories and their corresponding NFRs.

As shown in above Table 20, overall the ‘accuracy’ was considered as top ranked NFR in agile software development. Agile software development is an approach which attempt to hit the right targets which is possible only if the overall features, stories and task are being achieved with utmost accuracy. The accuracy in agile software development is such an important NFR which decrease the number of revisions of user stories and lead to agility. Secondly, in a case when changes happen it makes easy for agile team and client to measure them accurately and proceed quickly [8], [53]. Along with the accuracy of results, the second most important property to make an agile product successful is the delivery of

TABLE 11. Judgement matrix of external NFRs based on each expert response.

	Accessibility	Compatibility	Quality	Security
1 st Expert	Accessibility	1.00	4.00	3.00
	Compatibility	0.25	1.00	0.33
	Quality	0.33	3.00	1.00
	Security	0.14	0.33	0.20
2 nd Expert	Accessibility	1.00	3.00	7.00
	Compatibility	0.33	1.00	5.00
	Quality	0.14	0.20	1.00
	Security	0.11	0.14	0.33
3 rd Expert	Accessibility	1.00	7.00	3.00
	Compatibility	0.14	1.00	0.20
	Quality	0.33	5.00	1.00
	Security	0.20	3.00	0.33
4 th Expert	Accessibility	1.00	0.20	0.11
	Compatibility	5.00	1.00	1.00
	Quality	9.00	1.00	1.00
	Security	1.00	0.20	0.20
5 th Expert	Accessibility	1.00	0.14	0.20
	Compatibility	7.00	1.00	2.00
	Quality	5.00	0.50	1.00
	Security	3.00	0.33	0.33
6 th Expert	Accessibility	1.00	4.00	3.00
	Compatibility	0.25	1.00	0.50
	Quality	0.33	2.00	1.00
	Security	3.00	5.00	4.00
7 th Expert	Accessibility	1.00	7.00	1.00
	Compatibility	0.14	1.00	0.20
	Quality	1.00	5.00	1.00
	Security	0.33	3.00	0.14
8 th Expert	Accessibility	1.00	1.00	4.00
	Compatibility	1.00	1.00	4.00
	Quality	0.25	0.25	1.00
	Security	0.17	0.17	0.33
9 th Expert	Accessibility	1.00	5.00	4.00
	Compatibility	0.20	1.00	0.50
	Quality	0.25	2.00	1.00
	Security	0.14	0.33	0.33
10 th Expert	Accessibility	1.00	0.33	2.00
	Compatibility	3.00	1.00	4.00
	Quality	0.50	0.25	1.00
	Security	0.33	0.20	0.50
11 th Expert	Accessibility	1.00	7.00	1.00
	Compatibility	0.14	1.00	0.20
	Quality	1.00	5.00	1.00
	Security	0.33	3.00	0.14
12 th Expert	Accessibility	1.00	0.33	0.14
	Compatibility	3.00	1.00	0.33
	Quality	7.00	3.00	1.00
	Security	7.00	3.00	3.00

such result efficiently so that a good user experience could be imparted to users. So an appropriate level of efficiency should be defined keeping in view the criteria of effectivity and good

TABLE 12. Judgement matrix of process NFRs based on each expert response.

		Acceptance	Feasibility	Scalability
1 st Expert	Acceptance	1.00	4.00	5.00
	Feasibility	0.25	1.00	3.00
	Scalability	0.20	0.33	1.00
2 nd Expert	Acceptance	1.00	3.00	7.00
	Feasibility	0.33	1.00	5.00
	Scalability	0.14	0.20	1.00
3 rd Expert	Acceptance	1.00	4.00	5.00
	Feasibility	0.25	1.00	3.00
	Scalability	0.20	0.33	1.00
4 th Expert	Acceptance	1.00	0.33	5.00
	Feasibility	3.00	1.00	9.00
	Scalability	0.20	0.11	1.00
5 th Expert	Acceptance	1.00	2.00	5.00
	Feasibility	0.50	1.00	2.00
	Scalability	0.20	0.50	1.00
6 th Expert	Acceptance	1.00	0.33	5.00
	Feasibility	3.00	1.00	7.00
	Scalability	0.20	0.14	1.00
7 th Expert	Acceptance	1.00	5.00	7.00
	Feasibility	0.20	1.00	3.00
	Scalability	0.14	0.33	1.00
8 th Expert	Acceptance	1.00	0.50	0.25
	Feasibility	2.00	1.00	0.33
	Scalability	4.00	3.00	1.00
9 th Expert	Acceptance	1.00	1.00	7.00
	Feasibility	1.00	1.00	3.00
	Scalability	0.14	0.33	1.00
10 th Expert	Acceptance	1.00	5.00	7.00
	Feasibility	0.20	1.00	3.00
	Scalability	0.14	0.33	1.00
11 th Expert	Acceptance	1.00	4.00	0.25
	Feasibility	0.25	1.00	0.13
	Scalability	4.00	8.00	1.00
12 th Expert	Acceptance	1.00	0.25	0.17
	Feasibility	4.00	1.00	0.33
	Scalability	6.00	3.00	1.00

user experience. In current era, most of the application are being developed as web based system and interactive applications which should be accessible to the intended audience otherwise the product will be failed drastically. The security

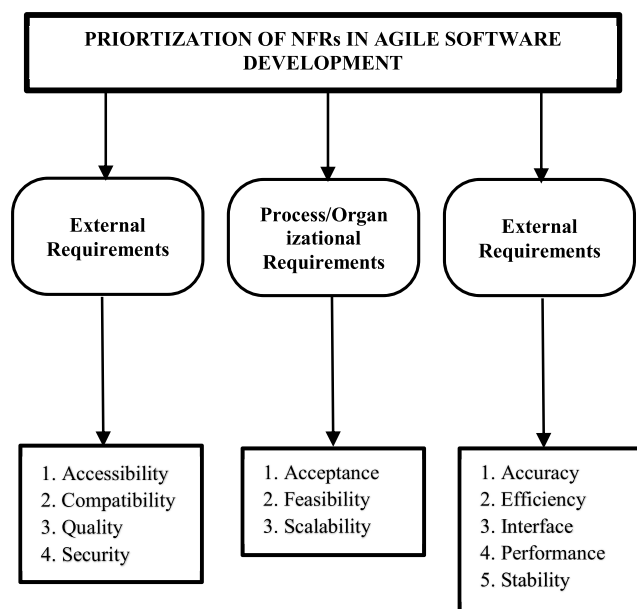
TABLE 13. Judgement matrix of product NFRs based on each expert response.

		Accuracy	Efficiency	Interface	Performance	Stability
1 st Expert	Accuracy	1.00	0.50	4.00	3.00	3.00
	Efficiency	2.00	1.00	7.00	5.00	5.00
	Interface	0.25	0.14	1.00	0.50	0.33
	Performance	0.33	0.20	2.00	1.00	2.00
	Stability	0.33	0.20	3.00	0.50	1.00
2 nd Expert	Accuracy	1.00	3.00	4.00	5.00	7.00
	Efficiency	3.00	1.00	2.00	2.00	4.00
	Interface	0.33	0.50	0.50	2.00	3.00
	Performance	0.25	0.50	0.50	1.00	3.00
	Stability	0.20	0.25	0.25	0.33	1.00
3 rd Expert	Accuracy	1.00	7.00	3.00	1.00	1.00
	Efficiency	0.14	1.00	0.14	0.20	0.20
	Interface	0.33	7.00	1.00	1.00	1.00
	Performance	1.00	5.00	1.00	1.00	1.00
	Stability	1.00	5.00	1.00	1.00	1.00
4 th Expert	Accuracy	1.00	3.00	4.00	5.00	7.00
	Efficiency	0.33	1.00	3.00	5.00	6.00
	Interface	0.25	0.33	1.00	2.00	5.00
	Performance	0.20	0.20	0.50	1.00	4.00
	Stability	0.14	0.17	0.20	0.25	1.00
5 th Expert	Accuracy	1.00	0.33	3.00	0.20	1.00
	Efficiency	3.00	1.00	5.00	1.00	3.00
	Interface	0.33	0.20	1.00	0.20	1.00
	Performance	5.00	1.00	5.00	1.00	5.00
	Stability	1.00	0.33	1.00	0.20	1.00
6 th Expert	Accuracy	1.00	0.20	0.33	0.20	0.14
	Efficiency	5.00	1.00	3.00	1.00	0.33
	Interface	3.00	0.33	1.00	0.33	0.20
	Performance	5.00	1.00	3.00	1.00	0.33
	Stability	7.00	3.00	5.00	3.00	1.00
7 th Expert	Accuracy	1.00	6.00	5.00	2.00	3.00
	Efficiency	0.17	1.00	0.25	0.33	0.50
	Interface	0.20	4.00	1.00	0.25	0.50
	Performance	0.50	3.00	4.00	1.00	3.00
	Stability	0.33	2.00	2.00	0.33	1.00
8 th Expert	Accuracy	1.00	0.33	3.00	4.00	5.00
	Efficiency	3.00	1.00	5.00	6.00	7.00
	Interface	0.33	0.20	1.00	2.00	4.00
	Performance	0.25	0.17	0.50	1.00	2.00
	Stability	0.20	0.14	0.25	0.50	1.00
9 th Expert	Accuracy	1.00	0.50	4.00	3.00	3.00
	Efficiency	2.00	1.00	7.00	5.00	5.00
	Interface	0.25	0.14	1.00	0.50	0.33
	Performance	0.33	0.20	2.00	1.00	1.00
	Stability	0.33	0.20	3.00	1.00	1.00
10 th Expert	Accuracy	1.00	0.50	4.00	3.00	3.00
	Efficiency	2.00	1.00	7.00	5.00	5.00
	Interface	0.25	0.14	1.00	0.50	0.33
	Performance	0.33	0.20	2.00	1.00	1.00
	Stability	0.33	0.20	3.00	1.00	1.00

is also the main concern in every kind of system to preserve the important data and information on short and long term basis for the smooth operation of the application. The security requirements need to be defined based on the user stories collected through client.

TABLE 13. (Continued.) Judgement matrix of product NFRs based on each expert response.

		Accuracy	Efficiency	Interface	Performance	Stability
		1.00	2.00	3.00	4.00	5.00
11 th Expert	Accuracy	1.00	2.00	3.00	4.00	5.00
	Efficiency	0.50	1.00	2.00	4.00	5.00
	Interface	0.33	0.50	1.00	3.00	2.00
	Performance	0.25	0.25	0.33	1.00	3.00
	Stability	0.20	0.20	0.50	0.33	1.00
		Accuracy	Efficiency	Interface	Performance	Stability
		1.00	0.50	0.50	2.00	1.00
12 th Expert	Accuracy	1.00	0.50	0.50	2.00	1.00
	Efficiency	2.00	1.00	0.33	2.00	4.00
	Interface	2.00	3.00	1.00	5.00	4.00
	Performance	0.50	0.50	0.20	1.00	0.50
	Stability	1.00	0.25	0.25	2.00	1.00

**FIGURE 7. Hierarchical structure of NFRs for AHP implementation.**

Furthermore, the agile team should have to establish that system being developed need to be compatible with which devices and related software applications for the effective implementation. And define the level of performance of the system in different scenarios. The level of acceptance is need to be define with respect to whole functionality, it is especially important in those cases where it is hard to put in practice on site client.

To ensure the usability of the system there is need to define the user interface requirement in terms of overall layout, appearance and aesthetics. The definition of stability and quality NFRs ensure the overall ability of the of the product to continue function over time without failure. The second last attribute is feasibility which indicate the needs to highlight the operation and implementation aspects of the system. The last but not least is the scalability NFR defines which functionality can be extended to support the growth in business operations and which functionality need to be discarded to shrink the operation of

software application to meet the changing needs of client organization.

The below Figure 8 graphically represent the NFRs prioritization model.

D. STAGE 4: EVALUATION OF PROPOSED MODEL BY CASE STUDY

The above NFRs prioritization model shown in Figure 8 has been evaluated through a case study which consists of three agile software projects accomplished recently in an Islamabad based software company namely DSI World Wide Consultancy Pvt. Ltd. The proposed NFRs prioritization model (Figure 3) intends to serve agile process with respect to defining NFRs specifically for medium scale agile projects mainly undertaken to implement information systems. It is therefore three mid-sized agile software projects including (i) Employees Attendance Management System (E&AMS) (ii) Learning Management System (LMS) (iii) Filling Station Software (FSS) have been selected to evaluate the proposed model. These projects were implemented by the same agile developers who were involved in this study as agile experts. The evaluation of proposed model has been done through the context of software developed and experience of agile experts. The evaluation predominantly relies on agile experts' opinion to determine the validity of the proposed NFR's prioritization model. The major purpose was to provide professional judgement on validity of the model. The comparison was made using prototype of the same three systems which were developed without considering the NFRs prioritization model proposed in this study for the identification of non-functional requirements. The evaluation has been done using qualitative (i.e. interview) method. In interview both the users and agile developers of the selected projects have been undertaken. The users involved in evaluation have used both of the systems developed considering NFRs prioritization model and systems developed without considering NFRs prioritization model. Similarly, the agile developers who involved in the development of both kind of systems were part of evaluation. The questions asked them about their perception on system's overall quality (i.e. user satisfaction) and time and effort required to define non-functional requirements in agile development. Overall, the objective of our evaluation was to determine the influence of NFRs prioritization model on user satisfaction and/or acceptance, time and effort. So this study has compare the performance with regards to three aspects including (i) user satisfaction (ii) time and (iii) effort. The project implemented based on the NFRs prioritization model shows overall improvement in terms of user satisfaction, time and effort required to produce NFRs in comparison to their counterpart projects which were implemented without considering NFRs prioritization models. The evaluation results showed that the users were more satisfied from the systems implemented based on NFRs prioritization model and their acceptance towards such system was higher in relation to their counterpart systems. The agile experts' feedback expressed

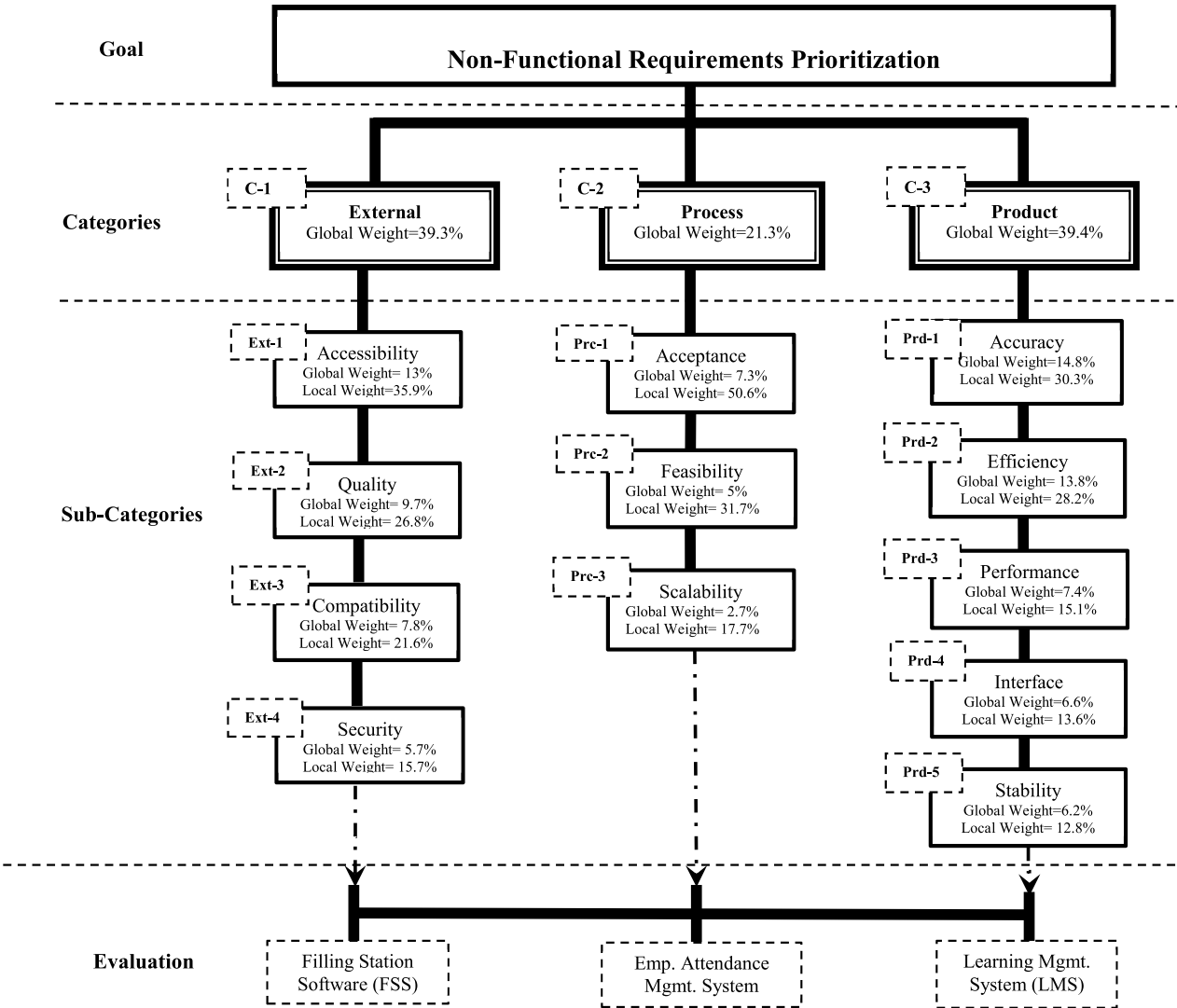


FIGURE 8. NFRs prioritization model.

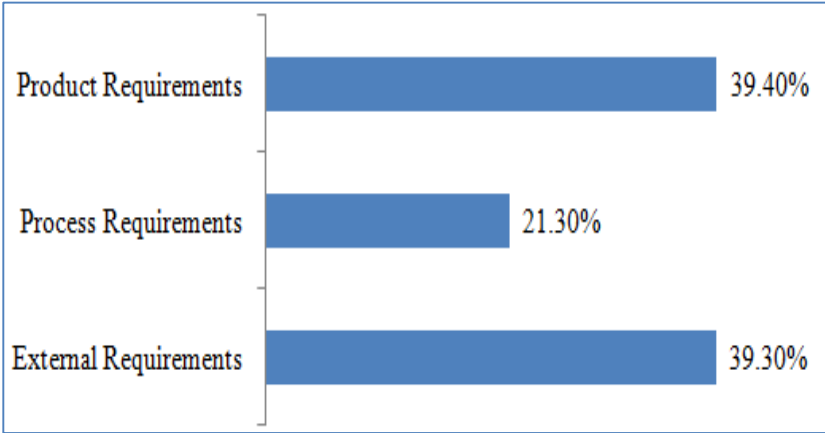


FIGURE 9. Graphical representation of prioritized major categories of NFRs.

that the proposed model helps requirement engineers to identify most promising non- functional aspects very quickly. It has also allowed better self-expression on the part of users. Furthermore, there is no extra effort required to produce

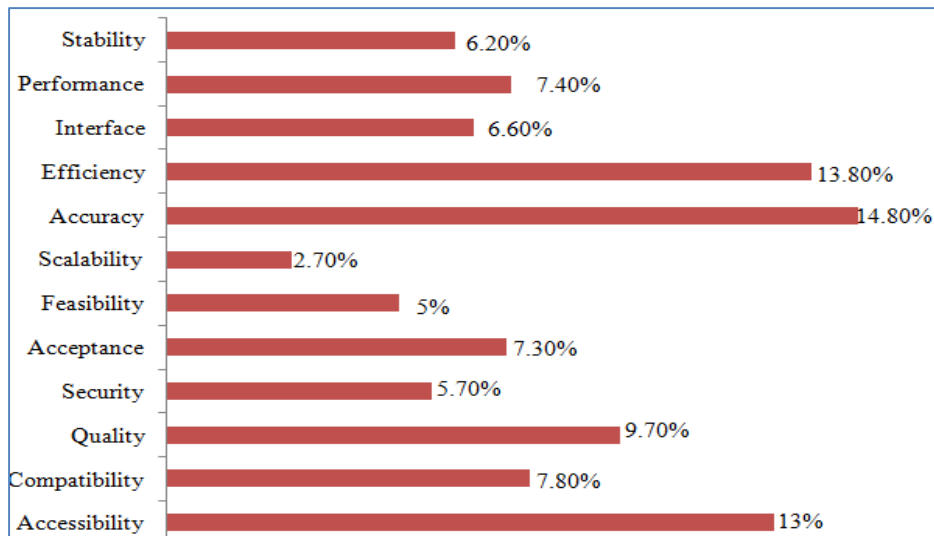


FIGURE 10. Graphical representation of prioritized NFRs of all categories.

TABLE 14. Overall aggregated priority weights of major categories of NFRs.

Major NFRs Categories	1 st Expert	2 nd Expert	3 rd Expert	4 th Expert	5 th Expert	6 th Expert	7 th Expert	8 th Expert	9 th Expert	10 th Expert	11 th Expert	12 th Expert
External NFRs	0.5571	0.6806	0.2299	0.7235	0.2674	0.6434	0.6327	0.6080	0.1149	0.2828	0.5794	0.1226
Process NFRs	0.1226	0.2014	0.1222	0.0833	0.0637	0.0738	0.1924	0.2721	0.7028	0.0738	0.1865	0.3202
Product NFRs	0.3202	0.1179	0.6479	0.1932	0.6689	0.2828	0.1749	0.1199	0.1822	0.6434	0.2341	0.5571
Consistency Check Values	RI	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58
	$\lambda_{\max}$	3.0183	3.0247	3.0037	3.0658	3.0292	3.0655	3.0092	3.0741	3.0542	3.0655	3.0183
	CI	0.0092	0.0124	0.0018	0.0329	0.0146	0.0328	0.0046	0.0371	0.0271	0.0328	0.0270
	CR	0.0158	0.0213	0.0032	0.0567	0.0252	0.0565	0.0079	0.0639	0.0467	0.0565	0.0158

TABLE 15. Overall priority weights of requirements after aggregation under each category.

External NFRs	1 st Expert	2 nd Expert	3 rd Expert	4 th Expert	5 th Expert	6 th Expert	7 th Expert	8 th Expert	9 th Expert	10 th Expert	11 th Expert	12 th Expert
Accessibility	0.5396	0.5739	0.5579	0.0693	0.2674	0.2728	0.3832	0.4078	0.6034	0.2333	0.3832	0.0530
Compatibility	0.1315	0.2913	0.0569	0.3946	0.0637	0.0772	0.0587	0.4078	0.1364	0.5423	0.0587	0.1404
Quality	0.2715	0.0903	0.2633	0.4571	0.6689	0.1243	0.4402	0.1254	0.1957	0.1397	0.4402	0.3014
Security	0.0573	0.0445	0.1219	0.0789	0.2674	0.5256	0.1178	0.0590	0.0646	0.0847	0.1178	0.5052
Consistency Check Values	RI	0.90	0.90	0.90	0.90	0.90	0.90	0.90	0.90	0.90	0.90	0.90
	$\lambda_{\max}$	4.1406	4.2092	4.1459	4.0460	3.0292	3.0292	4.2183	4.0786	4.1179	4.0602	4.2183
	CI	0.0469	0.0697	0.0486	0.0153	0.0146	0.0146	0.0728	0.0262	0.0393	0.0201	0.0728
	CR	0.0521	0.0775	0.0540	0.0170	0.0252	0.0252	0.0809	0.0291	0.0437	0.0223	0.0809

adequate NFRs which eventually reduce the cost. On the other hand, in the absence of proposed model there is an extra effort required on deciding an appropriate set of NFRs which ultimately lead to more cost.

V. CONCLUSION

Agile software development is an optimal alternative to implement software projects at rapid pace in comparison to traditional software development processes (Eberlein &

TABLE 16. Overall weights and ranking after aggregating main categories and NFRs.

Organizational or Process NFRs		1 st Expert	2 nd Expert	3 rd Expert	4 th Expert	5 th Expert	6 th Expert	7 th Expert	8 th Expert	9 th Expert	10 th Expert	11 th Expert	12 th Expert
Consistency Check Values	Acceptance	0.6651	0.6434	0.6651	0.2674	0.5949	0.2828	0.7235	0.1373	0.5105	0.7235	0.2267	0.0869
	Feasibility	0.2311	0.2828	0.2311	0.6689	0.2766	0.6434	0.1932	0.2395	0.3893	0.1932	0.0718	0.2737
	Scalability	0.1038	0.0738	0.1038	0.0637	0.1285	0.0738	0.0833	0.6232	0.1001	0.0833	0.7015	0.6393
	RI	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58	0.58
	$\lambda_{\max}$	3.0869	3.0655	3.0869	3.0292	3.0055	3.0655	3.0658	3.0183	3.0809	3.0658	3.0542	3.0540
	CI	0.0435	0.0328	0.0435	0.0146	0.0028	0.0328	0.0329	0.0092	0.0405	0.0329	0.0271	0.0270
CR		0.0750	0.0565	0.0750	0.0252	0.0048	0.0565	0.0567	0.0158	0.0698	0.0567	0.0467	0.0466

TABLE 17. Priority weights of major categories of NFRs based on each expert response.

Product NFRs		1 st Expert	2 nd Expert	3 rd Expert	4 th Expert	5 th Expert	6 th Expert	7 th Expert	8 th Expert	9 th Expert	10 th Expert	11 th Expert	12 th Expert
Consistency Check Values	Accuracy	0.2600	0.4948	0.3064	0.4598	0.1162	0.0436	0.4247	0.2514	1.3439	0.2623	0.4040	0.1460
	Efficiency	0.4706	0.2069	0.0399	0.2738	0.3260	0.2017	0.0613	0.4991	2.4246	0.4744	0.2803	0.2384
	Interface	0.0544	0.1426	0.2030	0.1368	0.0673	0.0888	0.1098	0.1279	0.2739	0.0545	0.1574	0.4229
	Performance	0.1154	0.1053	0.2253	0.0906	0.4010	0.2017	0.2743	0.0747	0.5001	0.0985	0.0980	0.0781
	Stability	0.0995	0.0504	0.2253	0.0390	0.0895	0.4641	0.1300	0.0469	0.5546	0.1103	0.0602	0.1146
	RI	1.12	1.12	1.12	1.12	1.12	1.12	1.12	1.12	1.12	1.12	1.12	1.12
Consistency Check Values	$\lambda_{\max}$	5.1582	5.1172	5.1688	5.3228	5.1463	5.1283	5.3276	5.1731	5.0729	5.0729	5.2428	5.2519
	CI	0.0396	0.0293	0.0422	0.0807	0.0366	0.0321	0.0819	0.0433	0.0182	0.0182	0.0607	0.0630
	CR	0.0353	0.0262	0.0377	0.0720	0.0327	0.0286	0.0731	0.0386	0.0163	0.0163	0.0542	0.0562

TABLE 18. Priority weights of external NFRs based on each expert response.

Dimension	Overall priority Weight
Product NFRs	0.394
External NFRs	0.393
Process NFRs	0.213

Leite, 2002). However, agile practices do not encompass elicitation process for NFRs. Researchers have introduced several techniques to address NFRs in the context of agile development but most of these techniques involve the client to elicit NFRs. The involvement of the client or end-user in elicitation process of NFRs require specialty in understanding and defining right NFRs for agile projects. To achieve quality in agile projects the inclusion of NFRs is necessary, to address this challenge, a prioritized model of NFRs is proposed in this study which is validated using case study consists of three different agile software projects. To develop a model, agile software development literature was extensively reviewed and consequently almost 77 those NFRs were extracted which have impact on the quality of agile software projects. Afterward, the qualifying mechanism was applied and resultantly out of 77 NFRs, 37 NFRs were qualified and classified into three categories in accordance to the classification model of Kotonya & Sommerville. Finally, after making pairwise comparison and calculating priority weights the most significant

TABLE 19. Priority weights of process nfrs based on each expert response.

Dimension	Requirements	Overall priority Weight
External	Accessibility	0.359
	Compatibility	0.216
	Quality	0.268
Process	Security	0.157
	Acceptance	0.506
	Feasibility	0.371
Product	Scalability	0.177
	Accuracy	0.303
	Efficiency	0.282
	Interface	0.136
	Performance	0.150
	Stability	0.128

NFRs in the context agile software development were identified.

The proposed model seems a better alternative for the challenge of NFRs in agile software development. The proposed model provided a prioritized list of 12 NFRs of three categories along with their importance in agile software projects. In the proposed, model the “Product” NFRs revealed as the most important category for agile software

TABLE 20. Priority weights of process NFRs based on each expert response.

Main Categories	Priority weight with contribution %	NFRs	Local weight with contribution%	Local Rank	Global weight with contribution%	Global Rank
External NFRs	0.393 (39.3%)	Accessibility	0.359 (35.9%)	1	0.130 (13%)	3
		Compatibility	0.216 (21.6%)	3	0.078 (7.8%)	5
		Quality	0.268 (26.8%)	2	0.097 (9.7%)	10
		Security	0.157 (15.7%)	4	0.057 (5.7%)	4
Process NFRs	0.213 (21.3%)	Acceptance	0.506 (50.6%)	1	0.073 (7.3%)	7
		Feasibility	0.317 (31.7%)	2	0.050 (5%)	11
		Scalability	0.177 (17.7%)	3	0.027 (2.7%)	12
		Accuracy	0.303 (30.3%)	1	0.148 (14.8%)	1
Product NFRs	0.394 (39.4%)	Efficiency	0.282 (28.2%)	2	0.138 (13.8%)	2
		Interface	0.136 (13.6%)	4	0.066 (6.6%)	8
		Performance	0.151 (15.1%)	3	0.074 (7.4%)	6
		Stability	0.128 (12.8%)	5	0.062 (6.2%)	9

projects with 39.40% priority whereas “Process” NFRs are the least important with 21.30% priority. Overall from three categories of NFRs are ‘accuracy’, ‘efficiency’, and ‘accessibility’ are the most important NFRs with 14.8%, 13.8% and 13% overall priority respectively.

The model can be used as a roadmap to address NFRs in every agile project to produce an effective and quality product. The model can be utilized in accordance to the situation of a project being developed in agile context. For example, if there is a requirement to consider only any of the global category of NFRs or some selected NFRs from all global categories. Similarly, if the project team lead feels a need to consider all of the NFRs of all global categories for a project the proposed model can be useful for such situation.

At present agile software development is prevalent among software development organizations. The striking feature of agile development is embracing changes frequently and give priority to users. However, owing to such feature of agile, NFRs are commonly not identified in the requirement elicitation process. They are time and again ignored at the start and even disregarded in the later stages of software development. Resultantly, a poor quality software produced. So the analysis and inclusion of NFRs in the software being developed is crucial and important for its success. But the dilemma of agile software development is that it does not provide a clear approach for addressing NFRs. Previous research studies have introduced several NFR handling framework which involve user to define NFRs who do not fully understand all NFRs they mainly focus on functional aspects of the system.

Moreover, agile methodology has minimal documentation which in turn causes traceability problems of NFRs provided by customer or users. To improve NFRs identification in agile development this study proposed a framework in the form of NFRs prioritized model which is developed through the involvement of experience agile practitioners. The proposed

model can facilitate and help the agile team members in addressing the NFRs. During validation of the model, the team members quickly identify the NFRs for selected systems and the usage of each NFR in these systems compared with the case of the absence of such model. In evaluation it is revealed that the model can assist the team members to define the non-functional aspects of the software very quickly. Furthermore, the agile practitioners were agreed that the model is useful in agile software development practice. To further assess the usefulness of the proposed model a quantitative empirical investigation will be conducted as future work.

To make the proposed model more sound, we are aimed to extend the study by involving agile practitioners in the analysis of NFRs from various software organizations located in different countries. The extended model will be verified using qualitative as well as quantitative methods. It is also in the authors future plan to compare the results obtained in this study with other MCDM methods such as fuzzy AHP and fuzzy TOPSIS to discover the best suitable results.

APPENDIX A

GRAPHICAL REPRESENTATION OF DATA ANALYZED REGARDING NFRs

See Figs. 2–6.

APPENDIX B

PRIORITY WEIGHT TABLES REGARDING MAJOR CATEGORIES AND THEIR NFRs

See Tables 14–17.

APPENDIX C

GRAPHICAL REPRESENTATION OF PRIORTIZED CATEGORIES AND THEIR NFRs

See Figs. 9 and 10.

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